NETFLIX CUSTOMER CHURN & ENGAGEMENT ANALYTICS

Project Report

SQL, Python and predictive modelling to identify churn drivers and quantify revenue at risk

Prepared by Arpita · November 2026 · Dataset: 5,000 customers × 14 attributes (synthetic practice dataset)

5,000	50.3%	$33.0K	$62.7K
customers analysed	overall churn rate	monthly revenue lost to churn	annual revenue at risk (337 customers)

1. Executive Summary

Half of the 5,000 subscribers in this dataset have churned (50.3%), costing about **$33,000 per month** in subscription revenue. This analysis identifies who is leaving, what separates them from loyal customers, and which active customers the retention team should contact first.

Key findings

- **Engagement is the dominant driver.** Churn falls from 93.6% in the lowest engagement quartile to 1.0% in the highest.
- **There is an inactivity cliff.** Customers who logged in within the last week churn at 13.4%; those inactive for 30+ days churn at 75.1%.
- **Plan and payment method matter.** Basic churns at 61.8% versus 43.7% for Premium. Crypto and Gift Card payers churn about 15 points more than card payers.
- **Multi-profile accounts are stickier.** Accounts with 4–5 profiles churn at 37.5–40.6%, versus about 58% for 1–3 profiles.
- **$62.7K per year is at risk.** 337 still-active customers have bottom-half engagement and have been inactive 15+ days; 92% of them are on Premium or Standard plans.

> **Bottom line:** retention effort should target low-engagement, inactive subscribers. Region, device, gender and genre have almost no effect on churn and are not useful targeting levers.

A Gradient Boosting model reaches a ROC-AUC of 0.998, but feature-group testing (Section 7) shows this reflects a very clean, synthetic engagement signal. The business value of the project lies in the segment-level insights and the retention target list rather than in the model score.

2. Business Problem and Objectives

Every cancelled subscription is lost recurring revenue, and winning a customer back costs more than keeping one. The retention team needs to know where to focus limited effort. This project had three objectives:

- **Identify** which customer segments (plan, payment method, engagement, account type) churn most.
- **Quantify** the revenue at risk from customers who are still subscribed but showing churn-like behaviour.
- **Prioritise** a concrete list of customers and actions for a retention campaign.

3. Data

3.1 Dataset overview

The dataset contains 5,000 customers and 14 attributes covering demographics, subscription, viewing behaviour and payment. The target variable is churned (1 = churned, 0 = retained); 2,515 customers (50.3%) churned.

Column	Type	Description
customer_id	string	Unique customer identifier (not used as a feature)
age	integer	Customer age (18–70)
gender	category	Male / Female / Other
subscription_type	category	Basic / Standard / Premium
watch_hours	float	Total hours watched
last_login_days	integer	Days since last login (0–60)
region	category	Six geographic regions
device	category	Primary viewing device (5 types)
monthly_fee	float	Monthly subscription cost ($8.99 / $13.99 / $17.99)
payment_method	category	Credit Card, Debit Card, PayPal, Crypto, Gift Card
number_of_profiles	integer	Profiles under the account (1–5)
avg_watch_time_per_day	float	Average hours watched per day
favorite_genre	category	Most-watched genre (7 genres)
churned	integer	Target: 1 = churned, 0 = retained

3.2 Data quality and cleaning

- **No missing values, no duplicate rows and no duplicate customer IDs** were found.
- **10 rows (0.2%) had avg_watch_time_per_day above 24 hours**, with a maximum of 98.4, which is physically impossible. All 10 are still-active customers who logged in within the last day, and the average equals total watch_hours in 9 of them (half in the tenth). This points to a division artefact (hours divided by only 1–2 days of activity).
- **Decision:** values were capped at 24 hours/day. The rule is applied in one place (src/data.py) so the SQL, EDA and model results stay consistent.

> **Data note:** this is a synthetic/public practice dataset. Churn rates and model scores are illustrative and would differ on real production data.

4. Methodology

Step	Approach	Tools / location
1. Clean & validate	Duplicates, missing values, IQR outliers, impossible values	Python, notebook §1
2. SQL analysis	12 queries: KPIs, segmentation, CTEs, NTILE quartiles, RANK within groups	SQLite, sql/churn_analysis.sql
3. Exploratory analysis	Segment, engagement and inactivity analysis; correlations; leakage check	Pandas, Seaborn, notebook
4. Retention target list	Rule-based at-risk definition validated against observed churn	SQL Q10–Q11, BI export
5. Predictive modelling	3 classifiers, stratified split, 5-fold CV, feature-group ablation	Scikit-learn, src/train.py
6. Deployment	REST scoring endpoint and interactive app	FastAPI, Streamlit

5. Findings

Overall, 50.3% of customers churned. The figure below summarises churn by plan, region and payment method; only plan and payment method differ meaningfully from the average.

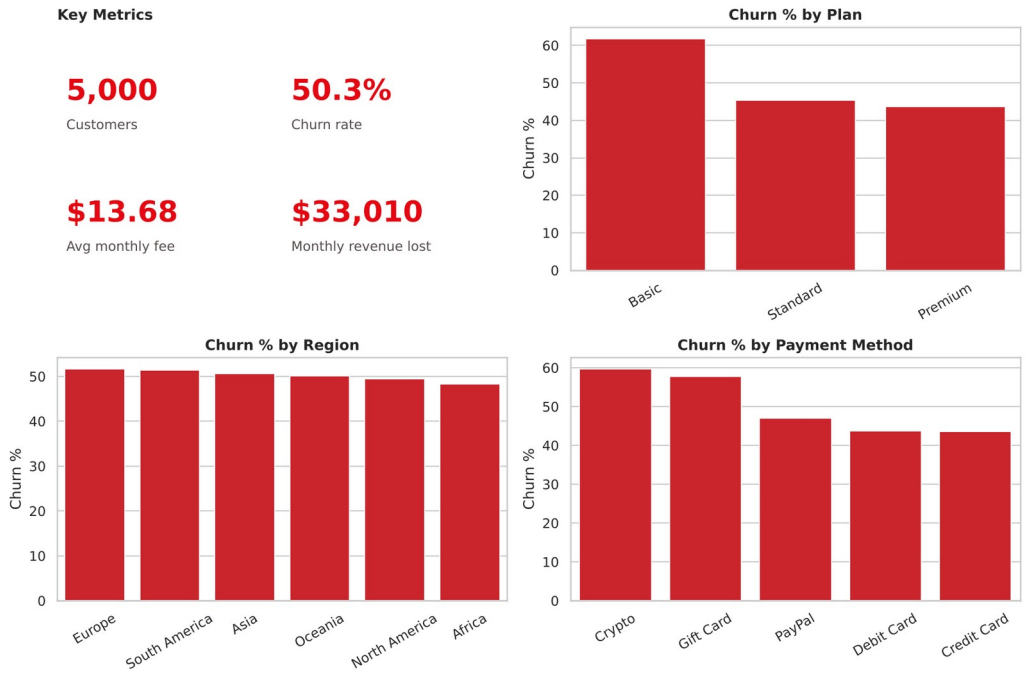


Figure 1. Key metrics and churn rate by plan, region and payment method

5.1 Subscription plan

Plan	Customers	Churn rate	Monthly revenue lost
Basic	1,661	61.8%	$9,233
Standard	1,646	45.4%	$10,465
Premium	1,693	43.7%	$13,313

Basic customers churn 18 points more than Premium. However, **Premium loses the most revenue** ($13.3K per month) because each customer pays more, so churn rate alone understates the Premium problem.

5.2 Payment method

Payment method	Customers	Churn rate
Crypto	995	59.7%
Gift Card	976	57.8%
PayPal	1,026	47.1%
Debit Card	1,030	43.7%
Credit Card	973	43.6%

Crypto and Gift Card payers churn about 15 points more than Credit/Debit Card payers. A plausible explanation is that these methods are prepaid or manual with no automatic renewal, although the dataset cannot confirm this.

5.3 Engagement

Customers were split into quartiles by average daily watch time. Engagement is by far the strongest separator of churners from loyal customers.

Engagement quartile	Hours per day	Customers	Churn rate
Q1 (lowest)	0.00 – 0.11	1,250	93.6%
Q2	0.11 – 0.29	1,250	74.4%
Q3	0.29 – 0.72	1,250	32.2%
Q4 (highest)	0.72 and above	1,250	1.0%

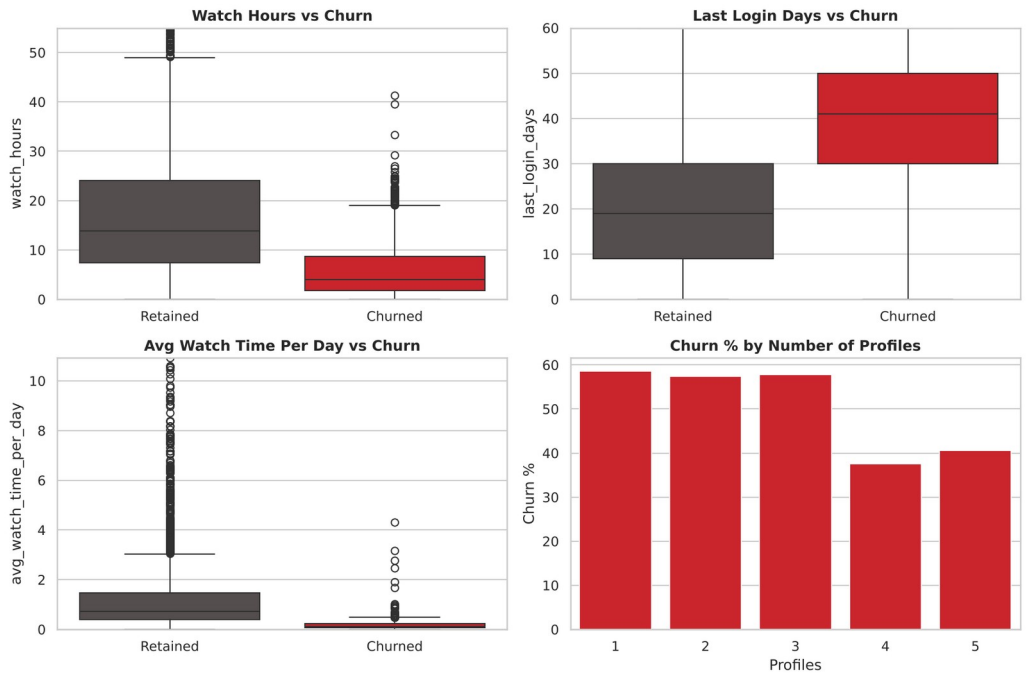


Figure 2. Viewing behaviour of retained vs churned customers, and churn by number of profiles

5.4 Inactivity

Days since last login	Customers	Churn rate
0 – 7 days	650	13.4%
8 – 14 days	559	27.0%
15 – 30 days	1,322	32.0%
30+ days	2,469	75.1%

Churn stays moderate up to 30 days of inactivity and then jumps to 75.1%. A re-engagement message at around day 15 would reach customers **before** the 30-day cliff.

5.5 Number of profiles

Profiles on account	Customers	Churn rate
1	972	58.6%
2	1,001	57.4%
3	994	57.8%
4	999	37.5%
5	1,034	40.6%

Accounts with 4–5 profiles churn roughly 18–20 points less than accounts with 1–3 profiles, suggesting that shared or household accounts are stickier.

5.6 What does not drive churn

Several attributes show almost no relationship with churn, with every group within about three points of the 50.3% average:

Attribute	Range of churn rates across groups
Region (6 groups)	48.3% – 51.7%
Device (5 groups)	49.2% – 51.8%
Gender (3 groups)	49.8% – 51.1%
Favourite genre (7 groups)	47.1% – 52.4%

Age shows a mild bump for the 45–54 group (54.0%) but no strong pattern (range 48.6%–54.0%). Knowing what does **not** matter is useful: the business should not spend targeting effort on these segments.

6. Revenue at Risk and Retention Target List

To turn the findings into an actionable list, an **at-risk customer** was defined as someone who:

- is **still subscribed** (churned = 0),
- has **bottom-half engagement** (quartiles 1–2, at most 0.29 hours per day), and
- has been **inactive for 15+ days**.

The definition is evidence-based: across the full customer base, customers matching this profile (2,281 of them) churned at **85.2%**. Of those, 337 have not churned yet. They are the ones a retention campaign can still save.

Plan	At-risk customers	Monthly revenue	Annual revenue
Premium	162	$2,914	$34,973
Standard	147	$2,057	$24,678
Basic	28	$252	$3,021
Total	**337**	**$5,223**	**$62,672**

Because Premium and Standard customers pay more, they account for 92% of the revenue at risk even though Basic has the highest churn rate. The full customer list is produced by SQL query Q10 and by the at_risk_flag column in the Power BI export file.

7. Predictive Modelling

7.1 Setup

Three classifiers were trained on a stratified 80/20 split (4,000 training rows, 1,000 test rows) using one-hot encoding for categorical features and standardisation for numeric features. Each was also evaluated with 5-fold cross-validation on the training set.

7.2 Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC	CV AUC
Logistic Regression	0.887	0.875	0.905	0.890	0.967	0.967
Random Forest	0.984	0.986	0.982	0.984	0.998	0.997

Model	Accuracy	Precision	Recall	F1	ROC-AUC	CV AUC
Gradient Boosting (best)	**0.988**	**0.990**	**0.986**	**0.988**	**0.998**	**0.999**

7.3 Is the score too good to be true?

Scores this high are unusual for real churn problems, so the result was stress-tested rather than taken at face value:

Check	Result
Single-feature ROC-AUC: avg_watch_time_per_day	0.924 (suspiciously high on its own)
Single-feature ROC-AUC: watch_hours / last_login_days	0.813 / 0.772
Relationship between the two watch features	avg_watch_time_per_day ≈ watch_hours ÷ days (implied 3–59 days for 90% of rows), so they are redundant
Model without watch_hours (top feature)	ROC-AUC 0.993 (almost unchanged)
Model without avg_watch_time_per_day only	ROC-AUC about 0.998 (unchanged)
Model without ALL three engagement features	ROC-AUC 0.628

> **Conclusion:** removing one engagement feature changes almost nothing because the others cover for it, but removing the whole group drops ROC-AUC from 0.998 to 0.63. Churn in this dataset is driven almost entirely by engagement, and the near-perfect score reflects a clean, synthetic signal. On real data, lower scores should be expected, and validation should use a time-based split.

Permutation importance (drop in ROC-AUC when a feature is shuffled) ranks watch_hours (0.098), avg_watch_time_per_day (0.095) and last_login_days (0.073) first, followed by number_of_profiles (0.032), payment_method (0.012), subscription_type and monthly_fee (0.005 each). Region, device, genre, gender and age contribute essentially nothing, which agrees with the segment analysis.

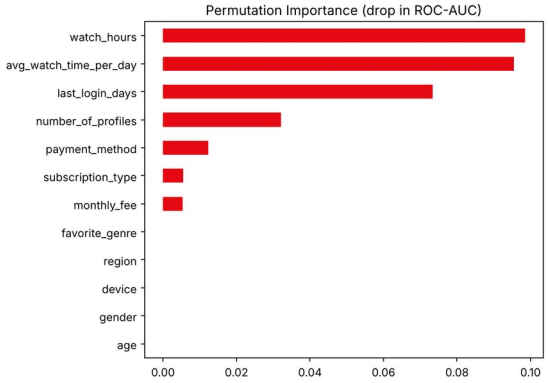


Figure 3. Permutation importance by feature

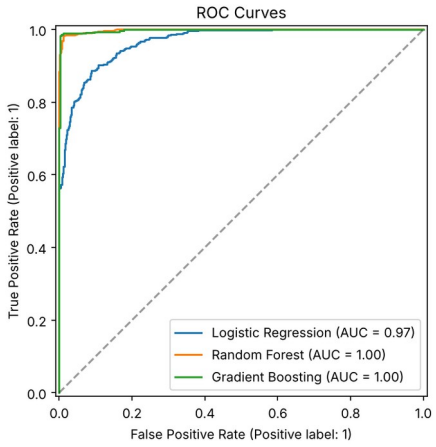


Figure 4. ROC curves for the three models

8. Recommendations

1. **Run a win-back campaign for the 337 at-risk customers**, prioritised by monthly fee, with a randomly selected control group so that the uplift can be measured.
2. **Trigger re-engagement at day 15 of inactivity**, before the 30-day point where churn jumps from 32% to 75%.
3. **Offer upgrade or bundle deals to Basic-plan users**, the highest-churn plan (61.8%).
4. **Move Crypto and Gift Card payers to auto-renewing card payment** with a small incentive; they churn about 15 points more than card payers.
5. **Encourage household use and extra profiles**: accounts with 4–5 profiles churn 18–20 points less.
6. **Prioritise Premium retention by revenue**, not churn rate: Premium has the largest revenue loss ($13.3K per month).
7. **Track the share of low-engagement customers weekly** as an early-warning KPI, since engagement is the strongest predictor of churn.

Measuring success: compare churn rate and retained revenue between treated and control customers, and net the retained revenue against the cost of the offer to estimate campaign ROI.

9. Limitations and Next Steps

Limitations

- The data is synthetic, so there is no real seasonality, customer history or business context.
- There are no signup or cancellation dates, so cohort retention and survival analysis were not possible.
- There is no cost data for retention offers, so campaign ROI could not be estimated.
- Engagement is measured alongside churn, so the model is descriptive; a true early-warning system needs time-stamped history.

Next steps

- Design and run an A/B test of the win-back offer.
- Estimate customer lifetime value to prioritise retention spend.
- Add signup and cancellation dates for cohort retention curves.
- Validate the model with a time-based train/test split.

Appendix: Reproducing the Analysis

All code, queries and the notebook are in the project repository. From the repository root:

```
pip install -r requirements.txt
python sql/load_data.py                          # build netflix.db
sqlite3 netflix.db < sql/churn_analysis.sql      # run the 12 SQL queries
python -m src.eda                                # EDA charts -> images/
python -m src.train                              # model + metrics -> models/
python -m src.export_for_bi                      # enriched CSV for Power BI
streamlit run src/frontend.py                    # interactive app
```

Repository: github.com/arpita7819/Netflix-Customer-Churn-Engagement-Analytics